

Target White Space Detection Engine

broken into 9 layers:

1. What “white space” really means for bags
 2. The data inputs you need
 3. The tagging engine design
 4. Mission tagging logic
 5. Cue tagging logic
 6. Size tagging logic
 7. Trend marker tagging logic
 8. Mission density computation and comparison
 9. Action signal generation and prioritization
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1) What “white space” means for bags

White space is not “they have SKUs and we don’t.”

White space is:

A specific bag mission in a specific price band where competitors are actively investing and Target is under-invested or mis-positioned, and where you can realistically win with feasible product concepts.

So white space must be defined at this grain:

Target Brand x Category x Mission x Price Band x Cue Profile

Example cell:

A New Day x Shoulder Bags x Occasion Glam x \$25–\$35 x Craft-forward

If this cell is strong in competitors and weak in Target then you have a potential white space.

2) Inputs the engine needs

You cannot build this on raw scrape fields alone. You need a normalized "feature record" per SKU per day.

Minimum fields per SKU snapshot:

- brand
- product_name
- category breadcrumbs
- price
- sale price or discount (optional)
- description or feature bullets (if available)
- material text
- dimensions (if available)
- image url
- color count (optional)
- timestamp
- product url

Then add:

- Target assortment table in the same normalized format (even if manual for v1)

This makes competitor and Target comparable.

3) Tagging engine design

This is the most important mindset:

Your tagging engine is not one classifier.

It is a layered inference pipeline.

Do NOT try to do "one LLM call" to label everything. You will get inconsistency.

Instead use a hybrid system:

- Deterministic rules first (high precision)
- Model inference second (coverage for edge cases)
- Human override third (quality control)

Tagging pipeline per SKU

For each SKU record:

1. Normalize text fields
2. Extract features
3. Apply rule based tags
4. Apply ML or LLM fallback tags with confidence
5. Resolve conflicts using priority rules
6. Store final tags + confidence + evidence tokens

You want every tag to be auditable.

Target leadership will ask "why did you call this 'travel utility'?"

You need an answer.

4) Mission tagging logic in brutal detail

Mission tags are "use case" buckets.

Here is a practical mission taxonomy for bags:

Mission tags:

- occasion_glam
- everyday_crossbody
- work_tote
- weekend_bucket
- travel_utility
- micro_mini_impulse
- slg_small_leathers
- gifting_novelty
- athleisure_active
- tech_carry

You can start with the first 7 and expand later.

Mission tagging is a scoring function, not a single label

Each SKU gets a mission score vector, then the top mission wins unless rules override.

Example:

SKU: "Beaded Mini Shoulder Bag"

- occasion_glam 0.85
- micro_mini_impulse 0.72
- everyday_crossbody 0.10

Final: occasion_glam

Secondary: micro_mini_impulse

You store both.

Mission scoring features

Define a feature dictionary.

Feature sources:

- Product name tokens
- Category breadcrumb tokens
- Description tokens
- Material tokens
- Size tokens
- Strap tokens
- Closure tokens

Example features for missions

Occasion glam signals:

- Tokens: party, evening, occasion, glam, sparkle, beaded, sequin, rhinestone, metallic, satin, clutch
- Shapes: clutch, minaudière, mini shoulder
- Colors: gold, silver, black, champagne
- Materials: satin, metallic PU, sequins

Everyday crossbody signals:

- Tokens: crossbody, messenger, hands-free, everyday, daily, casual
- Strap: adjustable strap, long strap
- Closure: zip top, secure
- Shape: camera bag, saddle, pouch crossbody

Work tote signals:

- Tokens: tote, work, office, laptop, commuter, carryall
- Size: medium-large
- Features: compartments, laptop sleeve, structured, top zip
- Handles: double handles, shoulder drop

Weekend bucket signals:

- Tokens: bucket, slouchy, hobo, relaxed
- Materials: woven, raffia, crochet, canvas
- Closure: drawstring

Travel utility signals:

- Tokens: travel, airport, nylon, water-resistant, anti-theft, secure, passport
- Closure: zip, hidden pocket
- Strap: crossbody, belt bag
- Materials: nylon, recycled poly

Micro mini impulse signals:

- Tokens: mini, micro, tiny, petite
- Price: typically lower
- Shape: mini shoulder, mini bag
- Trend markers: charms, novelty

SLG signals:

- Tokens: wallet, cardholder, coin purse, wristlet, phone pouch, organizer
- Category: small accessories

Rules to prevent dumb mislabels

Rule examples:

- If category indicates Wallets then mission must include slg_small_leathers regardless of name
- If token "clutch" present then occasion_glam gets a strong boost
- If token "laptop" present then work_tote wins even if "fashion tote" is in name
- If mini and beaded both present then dual mission: occasion_glam and micro_mini_impulse

Confidence computation

Confidence is based on:

- number of matched signals
- strength weights
- agreement across sources (name + category + description)

Confidence formula:

$\text{confidence} = 1 - \exp(-k * \text{signal_strength_sum})$

Where k is tuned so that 3 strong signals yields >0.8.

5) Cue tagging logic

Cue tags describe "why it sells" and "what it signals."

Cue types:

- craft_cue
- utility_cue
- luxury_cue
- sustainability_cue
- value_basic_cue
- trend_fast_cue

A SKU can have multiple cues, each with a weight.

Cue signals

Craft cue:

- beaded, embroidered, appliqué, crochet, woven, raffia, macramé, handwork, mirror work, patchwork
- textured surfaces and artisanal language

Utility cue:

- pockets, compartments, organizer, laptop sleeve, water-resistant, wipe-clean, secure zip, lightweight
- travel oriented language

Luxury cue:

- structured, minimal, leather, suede, quilted, croc, premium hardware, chain strap
- “polished” “elevated” language

Sustainability cue:

- recycled, vegan, eco, responsible, sustainable materials

Value basic cue:

- simple, classic, everyday essential, basic tote, multipack wallet

Trend fast cue:

- metallic, Y2K, baguette, cherry, charms, logo mania, seasonal colors

Cue vector

You compute a vector per SKU:

craft 0–1, utility 0–1, luxury 0–1, sustainability 0–1, trend 0–1, value 0–1

This is crucial because your white space engine uses cue mismatch between competitors and Target.

6) Size tagging logic

Size is tricky because dimensions are often missing.

So you must infer from tokens and category.

Size bands:

- micro

- mini
- small
- medium
- large
- oversized

Size inference layers

Layer 1: Explicit dimensions

If available, use actual:

- width, height, depth

Then map to size band thresholds.

Example thresholds for handbags:

- micro: width < 15 cm
- mini: 15–20 cm
- small: 20–26 cm
- medium: 26–34 cm
- large: 34–42 cm
- oversized: > 42 cm

Layer 2: Token inference

Tokens:

- micro, mini, small, large, XL, oversized

These override.

Layer 3: Category heuristic

- tote defaults to medium-large
- clutch defaults to micro-mini
- wallet defaults to micro
- backpack defaults to medium-large

Store size band with confidence.

7) Trend marker tagging logic

Trend markers are named signals that let you say:

"Competitors are leaning into X trend."

This is your deck's sexiest output.

Trend marker examples:

- metallic
- beadwork
- quilted
- woven_rafia
- croc_embossed
- chain_strap
- slouchy_hobo
- baguette
- bow_detail
- charm_bag
- logo_monogram
- contrast_stitch
- studded
- fringe
- denim_bag
- clear_pvc

How to tag trend markers

Trend tagging is essentially a keyword and pattern dictionary plus optional image inference later.

Rules:

- If token "metallic" or "foil" then metallic trend
- If token "beaded" or "rhinestone" then beadwork trend
- If token "quilted" then quilted trend

- If “chain strap” or “chain handle” then chain_strap trend

Make sure you store:

- trend_tag
- evidence tokens
- confidence

Optional Phase 2:

Image based detection for:

- quilted pattern
- metallic shine
- silhouette type

But ship with text first.

8) Mission density computation and comparison

This is the heart.

Step 1: Define your analysis cells

For each brand, on a given date:

Group SKUs by:

- category
- mission
- price_band

Optionally add:

- cue profile buckets like craft-high, utility-high etc.

Price bands example:

- 0_15
- 15_25
- 25_35
- 35_50
- 50_75

- 75_plus

Step 2: Compute mission density

For each brand and cell:

density = count of SKUs in cell / total SKUs in category

Example:

Zara has 200 handbags total

40 are occasion_glam in 25_35

Density = 20 percent

Compute for each competitor and for Target.

Step 3: Aggregate competitor density

You need a single competitor benchmark.

Use:

- median density across competitors
and
- max density across aspirational brands

So you can report:

- "Median competitor density"
- "Aspirational ceiling density"

This makes your argument more credible.

Step 4: Gap computation

Gap in that cell:

gap = competitor_median_density - target_density

If gap is positive and above threshold, it's a candidate.

Also check:

Target absolute count is below a minimum.

Because density can be misleading when totals are small.

So add:

- competitor_count_in_cell
 - target_count_in_cell
-

9) Action signal generation

This is where you turn analysis into “do this next.”

Action signals must include:

- opportunity type
- severity
- recommended action
- proof points
- competitor examples
- feasibility note

Basic rule to flag opportunity

If:

competitor_median_density > target_density

and price_band overlaps

and competitor_count_in_cell >= minimum_threshold

then flag.

Minimum threshold example:

- competitor_count_in_cell >= 8

This avoids noise.

Upgrade to a score based system

Compute an Action Score 0–100 using:

A) Demand signal

- competitor density
- newness rate
- momentum (net adds last 7 days)

B) Target gap

- target count shortfall
- mismatch in cue vector

C) Feasibility

- can Radnik produce in required cost band
- complexity risk

ActionScore = weighted sum

Then classify:

- 80+ Do now, sample 2 concepts
- 60–79 Investigate, build mini range
- 40–59 Monitor
- below 40 Ignore

What does an action signal look like in output

Action Signal object:

- target_brand: A New Day
- category: Shoulder bags
- mission: occasion_glam
- price_band: 25_35
- gap: +14 percent density vs competitors
- action_score: 86
- reason_codes:
 - competitor_density_high
 - competitor_momentum_positive
 - target_gap_large
 - craft_cue_missing
- top_examples:
 - Zara SKU link
 - Abercrombie SKU link

- recommended_concepts:
 - "Metallic Mini Shoulder with Chain Strap"
 - "Beaded Flap Mini Bag"
- feasibility_note:
 - "FOB target 5.5–7.0, hardware standardization recommended, bead shedding risk control"

This is what gets forwarded.

Implementation detail your team must not skip

Evidence tracking

Every auto tag needs to store:

- evidence tokens matched
- source fields used

So you can explain it.

Conflict resolution

If mission tags conflict:

- use priority rules based on category
- use highest confidence
- allow 2 missions if within 0.1 confidence difference

Versioning

Tag dictionaries change. You must version tag rules so past data remains reproducible.

Add:

- tag_rule_version to product_tag table

This is what makes the tool enterprise credible.

Why this engine wins Target

Because it shifts from:

"Here is what competitors sell"

to

"Here is what you should do next week, and why"

That is co-development value.
